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| • **NOISE—"CLUNK" NOISE HEARD ON 2-1 DOWNSHIFT VEHICLES EQUIPPED WITH 4R70W AUTOMATIC TRANSMISSION**<br>• **TRANSMISSION—4R70W—"CLUNK" NOISE HEARD FROM TRANSMISSION ON 2-1 DOWNSHIFT** | **Article No. 01-13-7** |

**FORD:** 1999-2001 CROWN VICTORIA, MUSTANG
1999 F-250 LD
1999-2001 E SERIES, EXPEDITION, EXPLORER, F-150

**LINCOLN:** 1999-2001 TOWN CAR

**MERCURY:** 1999-2001 GRAND MARQUIS, MOUNTAINEER

## ISSUE

Some vehicles equipped with a 4R70W automatic transmission may exhibit a "clunk" noise during the 2-1 downshift event. The "clunk" noise will be consistent in nature, and will occur during most 2-1 downshifts. This may be caused by a Planet Support Spring that is out of position.

## ACTION

Check for the correct installation of the Planet Support Spring. If the spring is not installed correctly, the Planetary Gear Support will make contact with the transmission case during the 2-1 downshift, causing the "clunk" noise. Refer to the following Service Procedure for details.

### SERVICE PROCEDURE

**NOTE**

**THE "CLUNK" NOISE DESCRIBED IN THIS ARTICLE SHOULD NOT BE CONFUSED WITH A HARSH/ROUGH 2-1 DOWNSHIFT CONDITION. IF A HARSH/ROUGH 2-1 SHIFT CONDITION IS PRESENT, FOLLOW NORMAL WORKSHOP MANUAL DIAGNOSTIC PROCEDURES.**

1. Use Chassis Ears (Rotunda number 107-R2102 or equivalent) to verify the "clunk" noise is originating from the transmission during the 2-1 downshift event.

**NOTE**

**IF THE NOISE IS NOT ORIGINATING FROM THE TRANSMISSION DO NOT CONTINUE WITH THIS SERVICE PROCEDURE. USE THE CHASSIS EARS TO HELP PINPOINT THE ORIGIN OF THE NOISE, THEN DIAGNOSE AND REPAIR AS NEEDED.**

2. Raise and support vehicle. Refer to Workshop Manual Section 100-02.
3. Check for correct torque on the following chassis & drivetrain components:
   - Driveshaft to Pinion Flange fasteners
   - Upper and Lower Control Arm fasteners (car applications)
   - Rear Leaf Spring Fasteners (truck applications)
   - Rear Shock Absorber fasteners
   - Transmission Mount and Crossmember fasteners
4. If any discrepancies are found, correct and test drive vehicle to verify repair. If condition is not corrected, continue to Step 5.
5. Remove the transmission from the vehicle and disassemble down to and including the Center Support Retaining Ring. Refer to Workshop Manual section 307-01.
6. Note the position of the Planet Support Spring. Compare to the correct installation which is illustrated in Figure 1. One tang of the Planet Support Spring should be loaded against the transmission case. The other tang should be loaded against the Planetary Gear Support.
7. Unless the Planet Support Spring is damaged, it may be reinstalled in the transmission correctly. If the spring is damaged, replace with part number F5AZ-7F277-AA.

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8. Install the Planet Support Spring in the position illustrated in Figure 1. Use a flat/wide tool, that will apply equal force across the spring. This will ensure the spring is installed squarely with the tangs loading the Transmission Case on one side and the Planetary Gear Support on the other.
9. Reassemble and install the transmission. Replace the Pump Gasket during reassembly. Part number F2VY-7A136-A.
10. Lower vehicle, refill fluid to correct level and verify repair.

| PART NUMBER | PART NAME |
|---|---|
| F5AZ-7F277-AA | Planet Support Spring |
| F2VY-7A136-A | Transmission Pump Gasket |

**OTHER APPLICABLE ARTICLES:** NONE
**WARRANTY STATUS:** Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage

| OPERATION | DESCRIPTION | TIME |
|---|---|---|
| 011307A | Check For The Correct Installation Of The Planet Support Spring (Includes Time To Remove And Install Transmission) 1999-2001 Explorer/Mountaineer 4X2 | 6.5 Hrs. |
| 011307B | Check For The Correct Installation Of The Planet Support Spring (Includes Time To Remove And Install Transmission And Transfer Case) 1999-2001 Explorer/Mountaineer 4X4 | 8.0 Hrs. |
| 011307C | Check For The Correct Installation Of The Planet Support Spring (Includes Time To Remove And Install Transmission) 1999-2001 Town Car, Grand Marquis, Crown Victoria | 5.9 Hrs. |
| 011307D | Check For The Correct Installation Of The Planet Support Spring (Includes Time To Remove And Install Transmission) 1999-2001 Econoline 4.2L 4.6L MFI 5.4L | 5.6 Hrs. |
| 011307E | Check For The Correct Installation Of The Planet Support Spring (Includes Time To Remove And Install Transmission) 1999-2001 F-150 4X2 4.6L MFI, 5.4L | 6.3 Hrs. |
| 011307F | Check For The Correct Installation Of The Planet Support Spring (Includes Time To Remove And Install Transmission) 1999-2001 F-150 4X2 4.2L | 6.4 Hrs. |
| 011307G | Check For The Correct Installation Of The Planet Support Spring (Includes Time To Remove And Install Transmission) 1999-2001 F-150 4X4 4.6L MFI, 5.4L | 8.1 Hrs. |
| 011307H | Check For The Correct Installation Of The Planet Support Spring (Includes Time To Remove And Install Transmission) 1999-2001 F-150 4X4 4.2L | 8.2 Hrs. |
| 011307I | Check For The Correct Installation Of The Planet Support Spring (Includes Time To Remove And Install Transmission) 1999-2001 Mustang 3.8L | 5.3 Hrs. |
| 011307K | Check For The Correct Installation Of The Planet Support Spring (Includes Time To Remove And Install Transmission) 1999-2001 Mustang 4.6L MFI | 5.7 Hrs. |
| 011307L | Check For The Correct Installation Of The Planet Support Spring (Includes Time To Remove And Install Transmission) 1999-2001 Expedition 4X2 | 5.9 Hrs. |
| 011307M | Check For The Correct Installation Of The Planet Support Spring (Includes Time To Remove And Install Transmission And Transfer Case) 1999-2001 Expedition 4X4 | 8.3 Hrs. |

**DEALER CODING**

| BASIC PART NO. | CONDITION CODE |
|---|---|
| 7F277 | 07 |

**OASIS CODES:** 502000, 504000, 597997

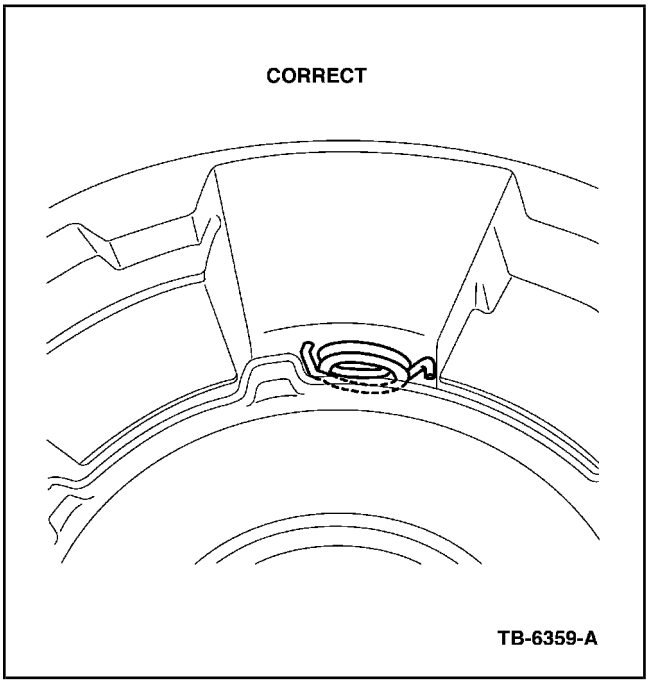


Figure 1 - Article 01-13-7

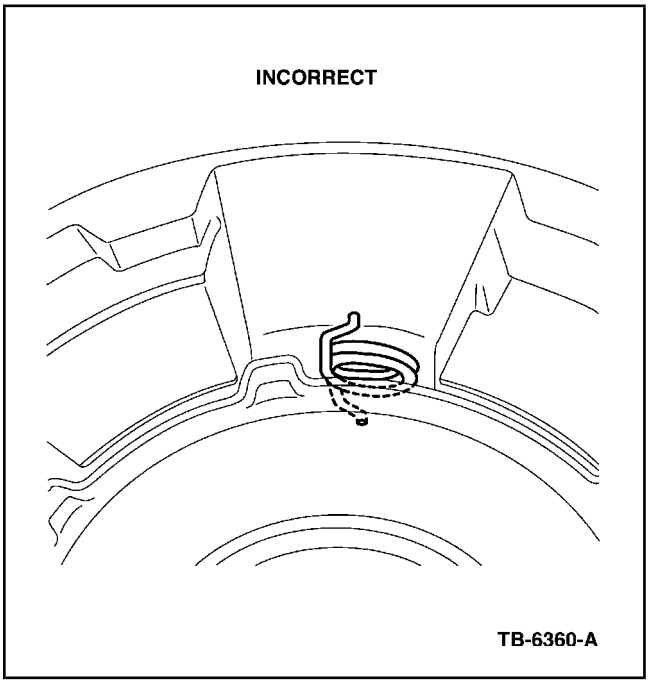


Figure 2 - Article 01-13-7